Forecasting stock returns under economic constraints

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Introduction

- Equity premium (EP) forecasts play a central role in areas in asset pricing.
- More than 25 years of research shows that models allowing for time-varying return predictability often produce worse out-of-sample forecasts.
- Economically motivated constraints offer the potential to sharpen forecasts, particularly when the data are noisy and parameter uncertainty is a concern as in return prediction models.

Introduction

- This paper proposes a new approach for incorporating economic information via inequality constraints on moments of the predictive distribution of the equity premium.
- Two types of economic constraints:
  1. the conditional mean of the equity premium to be nonnegative.
  2. the conditional Sharpe ratio (SR) has to lie between zero and a predetermined upper bound.

Introduction

- For first constraint: It is difficult to imagine an equilibrium setting in which risk averse investors would hold stocks if their expected compensations were negative, and so this seems like a mild restriction.
- For second: The zero lower bound is identical to the equity premium constraint, and the upper bound rules out that the price of risk becomes too high.

Model

$$r_{\tau+1} = \mu + \beta x_\tau + \epsilon_{\tau+1}, \tau = 1, ..., t-1,$$

$$\epsilon_{\tau+1} \sim N(0, \sigma_\epsilon^2)$$

where $r_{\tau+1}$ is stock returns in excess of a riskfree rate, x_τ is lagged predictor variables.

Equity premium constraint

- The parameters μ and β should be estimated subject to the constraint at all times:

$$\mu + \beta x_\tau \geq 0 \quad \text{for} \quad \tau = 1, ..., t$$

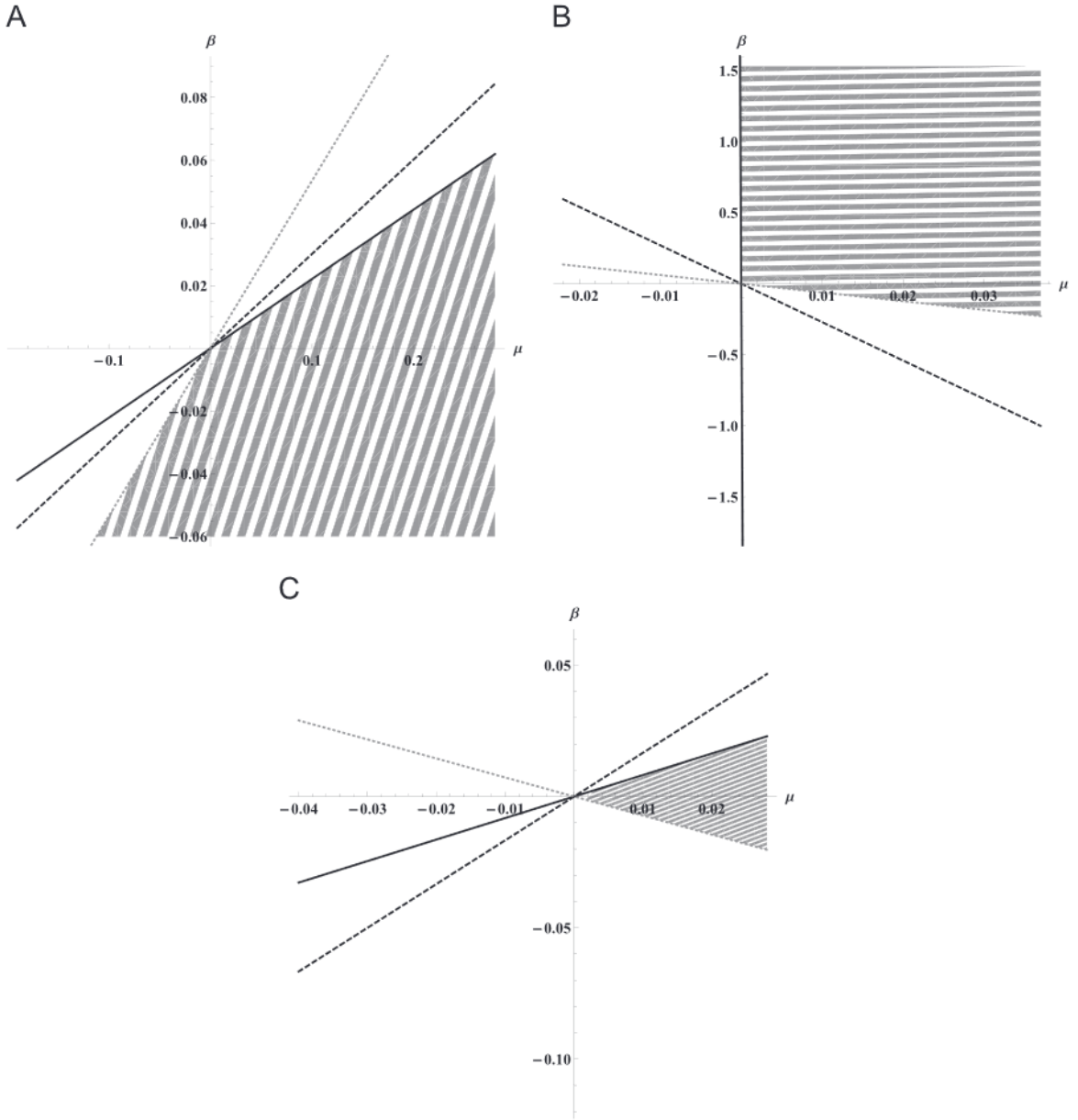

Fig. 1. Equity premium constraint. This figure shows the effect on the parameters μ (shown on the x-axis) and β (shown on the y-axis) from imposing the equity premium constraint $\mu+\beta x_t \geq 0$, as indicated by the shaded area. The solid and dotted lines correspond to the cases in which the predictor value is set to the smallest and largest values attained in sample, and the dashed line depicts the case in which the predictor value is set to its sample average. (a) Log dividend–price ratio, $-4.52 \leq x_t \leq -1.87$, (b) T-bill rate, $0.0001 \leq x_t \leq 0.163$, (c) Log dividend–payout ratio, $-1.22 \leq x_t \leq 1.38$.

Equity premium constraint

- Campbell and Thompson (2008) impose a truncated constrain

$$\hat{r}_{t+1|t} = max(0, \hat{\mu}_t + \hat{\beta}_t x_t) \tag{1}$$

This just simply overrule the forecast and do not impose on their parameters.

Sharpe ratio constraint

- For US data, the Sharpe ratio has been found to be time-varying and countercyclical
- More important, the empirical Sharpe ratio appears more volatile than what the leading asset pricing models would suggest.

Sharpe ratio constraint

- Modify equation 1 as:

$$r_{\tau+1} = \mu + \beta x_\tau + exp(h_{\tau+1})u_{\tau+1}$$

- where $h_{\tau+1}$ denotes the (log of) return volatility at time $\tau + 1$ and $u_{\tau+1} \sim N(0, 1)$. Following common stochastic volatility models, log volatility is assumed to evolve as a driftless random walk,

$$h_{\tau+1} = h_\tau + \xi_{\tau+1}$$

$$\xi_{\tau+1} \sim N(0, \sigma_\xi^2)$$

- u_τ and ξ_s are mutually independent for all τ and s.

Sharpe ratio constraint

- Next define the (approximate) annualized conditional Sharpe ratio at time τ as

$$SR_{\tau+1|\tau} = \frac{\sqrt{H}(\mu + \beta x_\tau)}{exp(h_\tau + 0.5\sigma_\xi^2)}$$

- where H denotes the number of observations per year (i.e., H = 12, 4; and 1 with monthly, quarterly, and annual data, respectively).

Sharpe ratio constraint

- We assume that the conditional Sharpe ratio is bounded both from below and above at all times:

$$SR^l \leq SR_{\tau+1|\tau} \leq SR^u \quad \text{for} \quad \tau = 1, ..., t$$

- In the empirical implementation, annualized values of $SR_{\tau+1|\tau}$ around 0.5 are seen as normal in the context of the market portfolio, given estimates of its mean and volatility. Sharpe ratios higher than one are highly improbable for a nonleveraged market portfolio, so we accordingly set $SR^u \leq 1$.

Priors

- Theoretical constraints are naturally interpreted as reflecting the forecaster's prior beliefs on return predictability.
- In this way, they can best be imposed using Bayesian techniques.
- A major advantage of our Bayesian approach is that we obtain the full predictive densities of returns in a way that accounts for parameter estimation error. Such densities are vastly more informative than point forecasts of excess returns based on conventional plug-in least squares estimates.

Priors

- Following standard practice, the priors for the parameters μ and β in Eq. (1) are assumed to be normal and independent of σ_ϵ^2,

$$\begin{bmatrix} \mu \\ \beta \end{bmatrix} \sim N(\underline{b}, \underline{V})$$

Priors

where

$$\underline{b} = \begin{bmatrix} \bar{r}_t \\ 0 \end{bmatrix}, \quad \underline{V} = \begin{bmatrix} \underline{\psi}^2 s_{r,t}^2 & 0 \\ 0 & \underline{\psi} s_{r,t}^2 / s_{x,t}^2 \end{bmatrix},$$

with data-based moments

$$\bar{r}_t = \frac{1}{t-1} \sum_{\tau=1}^{t-1} r_{\tau+1}, \quad s_{r,t}^2 = \frac{1}{t-2} \sum_{\tau=1}^{t-1} (r_{\tau+1} - \bar{r}_t)^2,$$

and

$$\bar{x}_t = \frac{1}{t-1} \sum_{\tau=1}^{t-1} x_\tau, \quad s_{x,t}^2 = \frac{1}{t-2} \sum_{\tau=1}^{t-1} (x_\tau - \bar{x}_t)^2.$$

Priors

- Here $\underline{\Psi}$ is a constant that controls the tightness of the prior, with $\underline{\Psi} \to \infty$ corresponding to a diffuse prior on μ and β.
- The terms $s_{r,t}^2$ and $s_{r,t}^2/s_{x,t}^2$ in the diagonal of the prior variance are scaling factors introduced to guarantee comparability of the priors across different predictors and across different data frequencies.

Priors

- Next, specify a gamma prior for the error precision of the return innovation σ_ϵ^{-2}

$$\sigma_\epsilon^{-2} \sim G(s_{r,t}^{-2}, \underline{\nu}_0(t-1))$$

- where $\underline{\nu}_0$ is a prior hyperparameter that controls the degree of informativeness of this prior, with $\underline{\nu}_0 \to 0$ corresponding to a diffuse prior

Priors

- The SR constraint equation (8) requires specifying a joint prior for the sequence of log return volatilities, h^t, and the error precision, σ_ϵ^{-2}. Writing $p(h^t, \sigma_\epsilon^{-2}) = p(h^t|\sigma_\epsilon^{-2})p(\sigma_\epsilon^{-2})$,it follows from Eq. (6) that

$$p(h^t|\sigma_\xi^{-2}) = \sum_{\tau=1}^{t-1} p(h_{\tau+1}|h_\tau, \sigma_\xi^{-2})p(h_1)$$

with $h_{\tau+1}|h_\tau, \sigma_\xi^{-2} \sim N(h_\tau, \sigma_\xi^{-2})$

Thus, we need to specify priors only for h_1, the initial log volatility, and σ_ξ^{-2}

Priors

Priors for h_1 and σ_ξ^{-2}:

$$h_1 \sim \mathrm{N}(\ln(s_{r,t}), \underline{k}_h)$$

and

$$\sigma_\xi^{-2} \sim G(1/\underline{k}_\xi, 1).$$

hyperparameters are choosen to be uninformative

Imposing economic constraints

- Starting with the EP constraint, we modify the priors on μ and β

$$\begin{bmatrix} \mu \\ \beta \end{bmatrix} \sim N(\underline{b}, \underline{V}), \quad \mu, \beta \in A_t$$

- where A_t is a set such that $A_t = \{\mu + \beta x_\tau \geq 0, \tau = 1, ..., t\}$
- Similarly, for the SR constraint, we restrict the priors on $\{\mu, \beta, \sigma_\xi^{-2}, h_1, h_2, ..., h_t\} \in \tilde{A}_t$, where $\tilde{A}_t$ is a set satisfying $\tilde{A}_t = \{SR^l \leq SR_{\tau+1|\tau} \leq SR^u, \tau = 1, ..., t\}$

Data

- Stock returns are computed from the Standard and Poor's (SP) 500 index and include dividends.
- 17 predictor variables originally analyzed in Welch and Goyal (2008) and subsequently extended up to 2010 by the same authors.
- A short T-bill rate is subtracted from stock returns to capture excess returns.

Data

- Univariate regressions with a single predictor variable
- Main analysis focus on three predictors: the log dividend–price ratio, the T-bill rate, and the default yield spread

Coefficient estimates and predictive densities

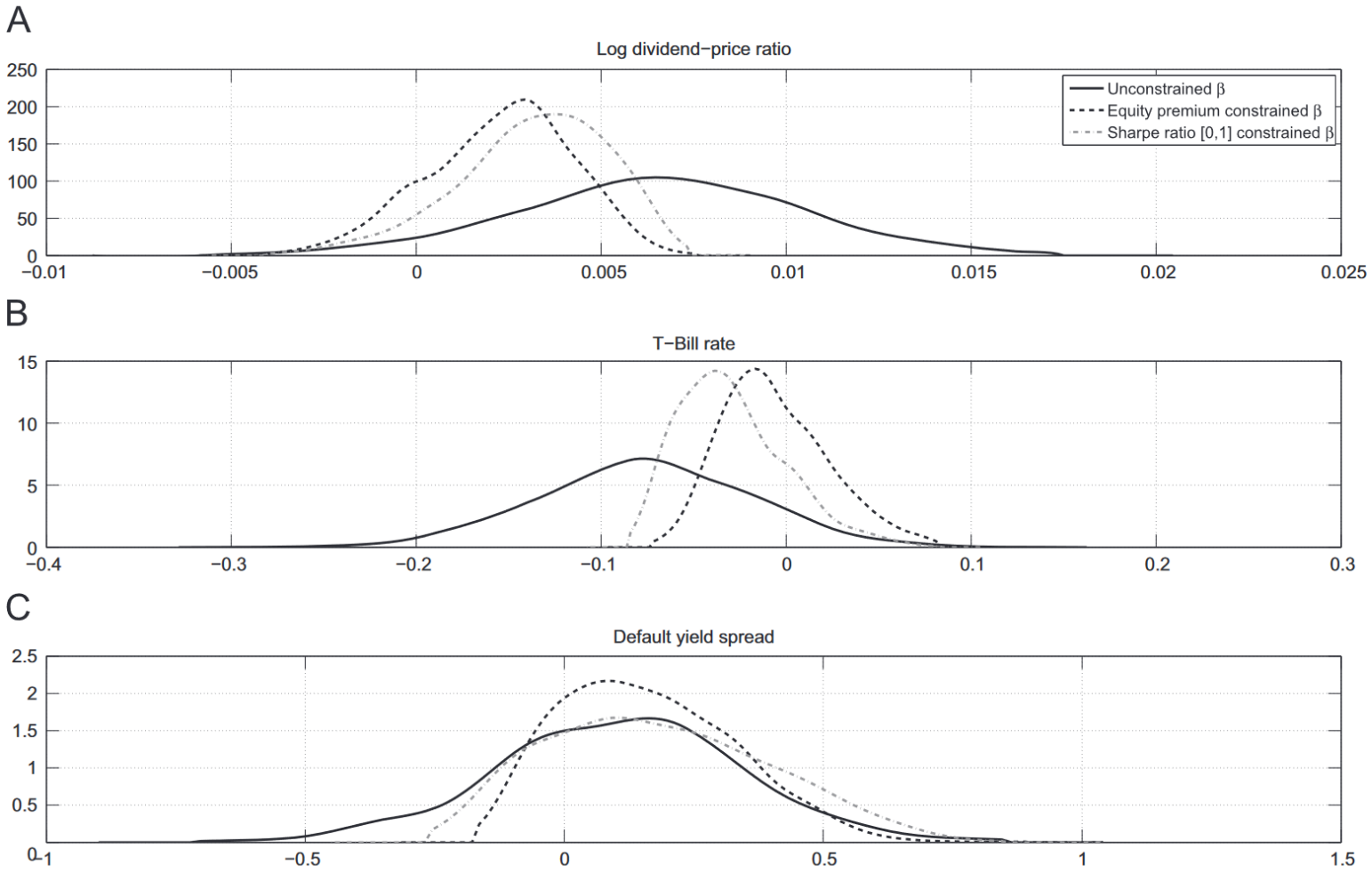

Fig. 4. Slope coefficient of predictors under constrained and unconstrained models. This figure shows the posterior density of the slope coefficient, β, from a regression of monthly excess returns (r_{t+1}) on an intercept and a lagged predictor variable, x_t: $r_{t+1} = \mu + \beta x_t + \varepsilon_{t+1}$. The equity premium constrained model imposes that $\hat{r}_{\tau+1|t} = \int(\mu+\beta x_\tau)p(\mu,\beta|\mathcal{D}^t)\,d\mu\,d\beta > 0$, for $\tau = 1,\ldots,t$ and information set $\mathcal{D}^t$, and the Sharpe ratio constraint imposes that $0 \le \hat{r}_{\tau+1|t}/\hat{\sigma}_{\tau+1|t} \le 1$, for $\tau = 1,\ldots,t$, where $\hat{\sigma}_{\tau+1|t}$ is the posterior volatility estimate obtained from a stochastic volatility model. The posterior density estimates shown here are based on the full sample at the end of 2010. Panels A, B, and C use the log dividend–price ratio, T-bill rate, and the default yield spread as the predictor, respectively.

Coefficient estimates and predictive densities

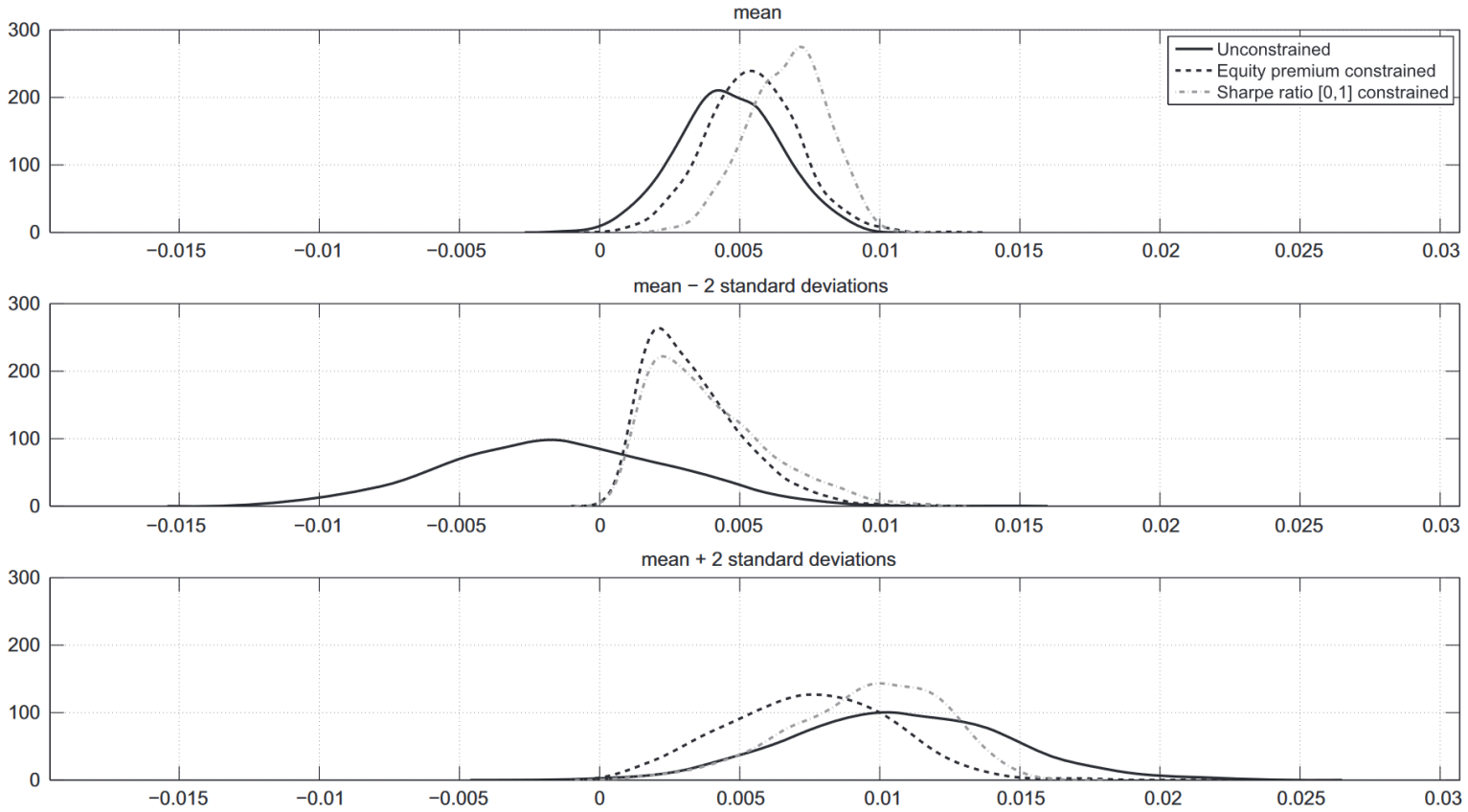

Fig. 5. Posterior density of the equity premium under constrained and unconstrained models (dividend–price ratio). This figure shows the posterior density of the equity premium as a function of the log dividend–price ratio, $\mu+\beta x_T$, where x_T is set at the sample mean of the log dividend–price ratio $\overline{x}$ (top), $\overline{x}-2st.dev(x)$ (middle), and $\overline{x}+2st.dev(x)$ (bottom). The equity premium constrained model imposes that $\hat{r}_{\tau+1|t}=\int(\mu+\beta x_\tau)p(\mu,\beta|\mathcal{D}^t)\,d\mu\,d\beta>0$, for $\tau=1,\ldots,t$ and information set $\mathcal{D}^t$, and the Sharpe ratio constraint imposes that $0\leq\hat{r}_{\tau+1|t}/\hat{\sigma}_{\tau+1|t}\leq 1$, for $\tau=1,\ldots,t$, where $\hat{\sigma}_{\tau+1|t}$ is the posterior volatility estimate obtained from a stochastic volatility model. All posterior density estimates are based on the full data sample as of the end of 2010.

Forecasts of equity premia

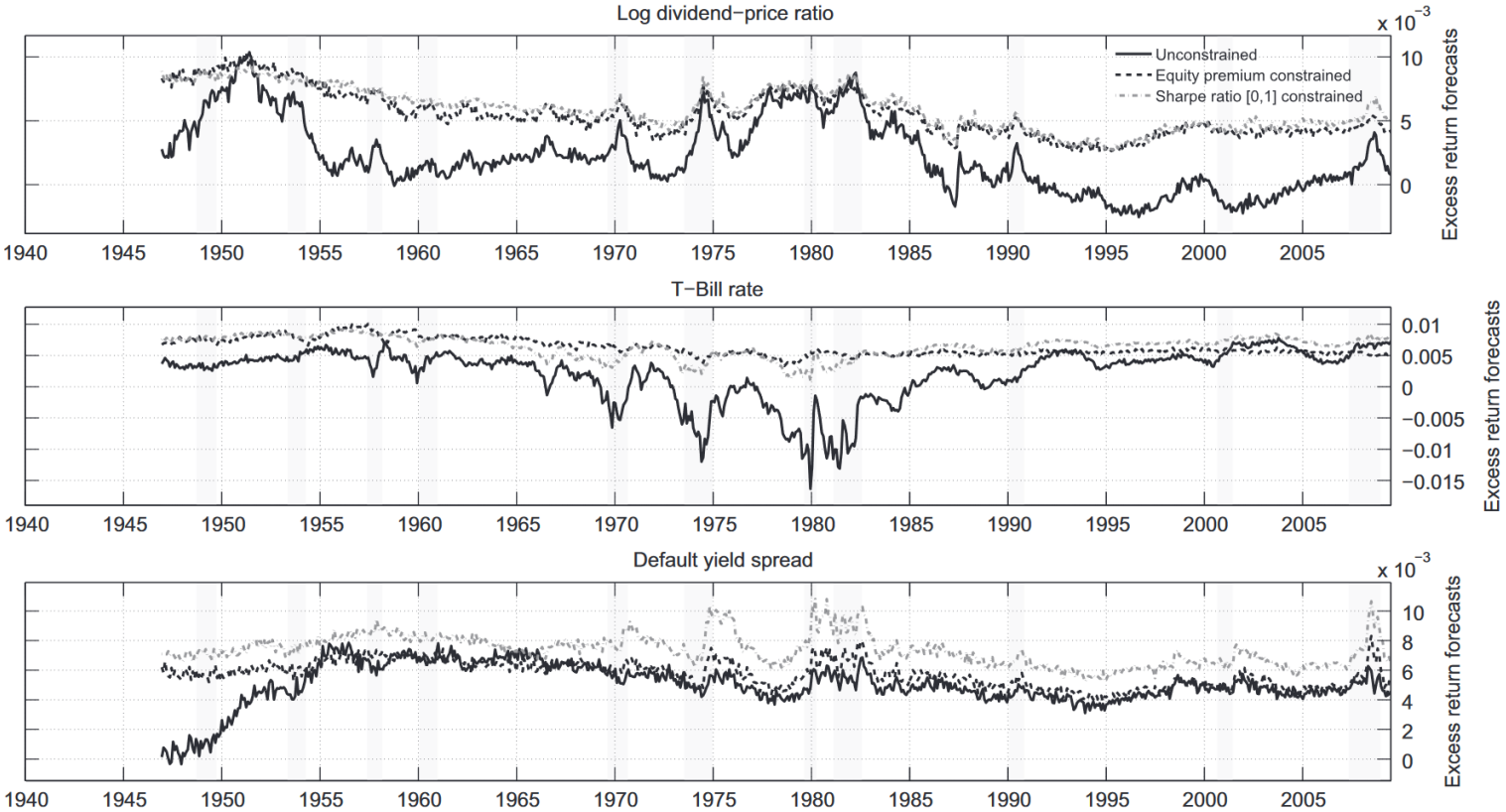

Fig. 9. Out-of-sample equity premium forecasts under unconstrained and constrained models. Each month we regress excess returns (r_{t+1}) on an intercept and a lagged predictor, x_t: $r_{t+1} = \mu + \beta x_t + \varepsilon_{t+1}$. Estimates from this model are then used to generate recursive one-step out-of-sample forecasts of excess returns and the process is repeated up to the end of the sample in 2010. The equity premium constrained model imposes that $\hat{r}_{\tau+1|t} = \int(\mu+\beta x_\tau)p(\mu,\beta|\mathcal{D}^t)\,d\mu\,d\beta > 0$, for $\tau = 1,\ldots,t$ and information set $\mathcal{D}^t$, and the Sharpe ratio constraint imposes that $0 \le \hat{r}_{\tau+1|t}/\hat{\sigma}_{\tau+1|t} \le 1$, $\tau = 1,\ldots,t$, where $\hat{\sigma}_{\tau+1|t}$ is the posterior volatility estimate obtained from a stochastic volatility model. The three graphs use different predictor variables: the log dividend–price ratio (top), the T-bill rate (middle), and the default yield spread (bottom).

Sharp Ratio

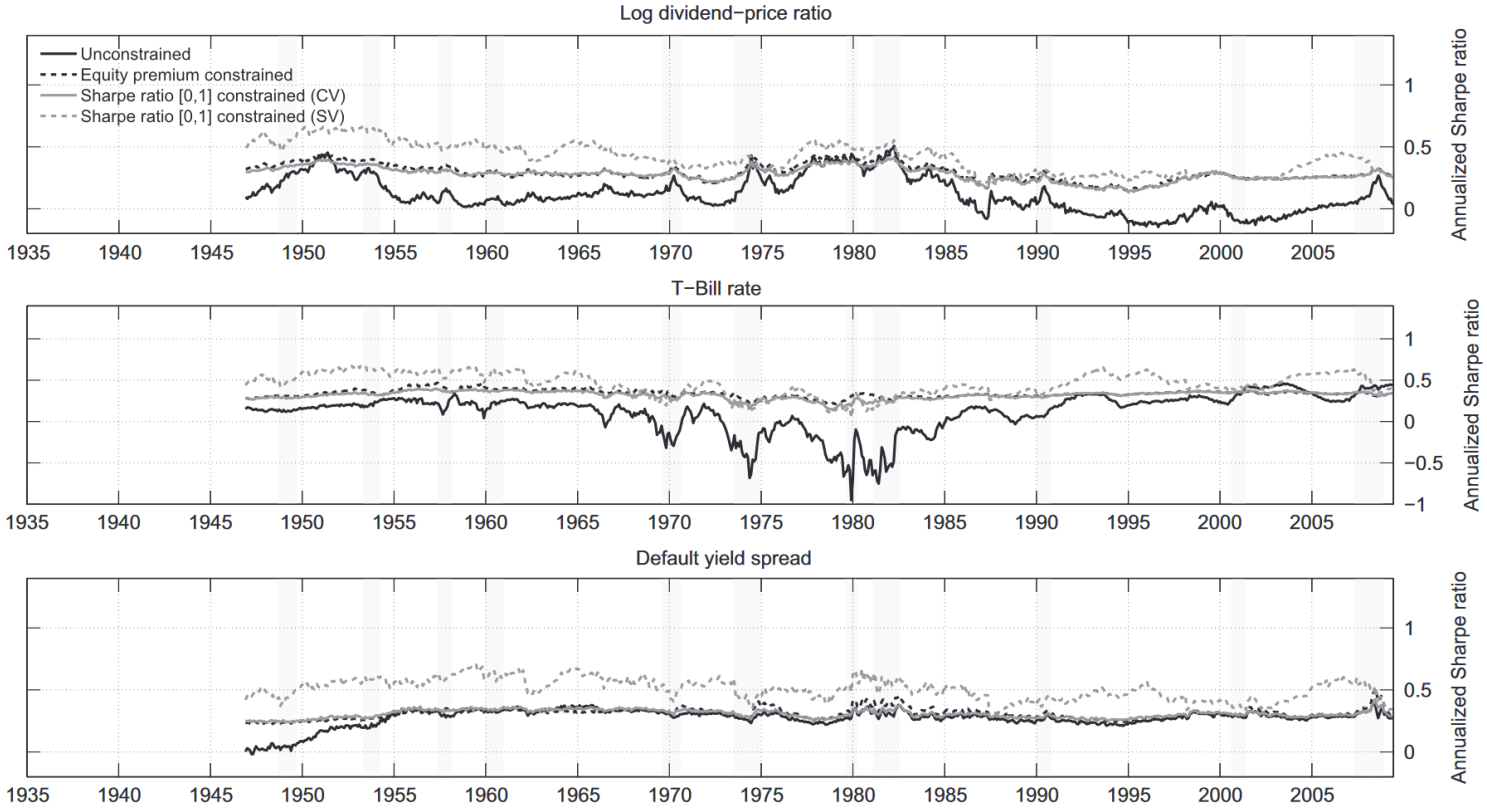

Fig. 11. Conditional Sharpe ratios. This figure shows the time series of conditional Sharpe ratios computed from the predictive density of excess returns based on the unconstrained, equity premium constrained, and Sharpe ratio constrained models [the last computed under both constant volatility (CV) and stochastic volatility (SV)]. The three graphs use different predictor variables: the log dividend–price ratio (top), the T-bill rate (middle), and the default yield spread (bottom).

Out-of-sample predictive performance (Monthly)

Variable	*Panel A: full sample (1947–2010)*			
	No constraint	CT truncation	EP constraint	SR [0,1] constraint
Log dividend–price ratio	0.10%*	**0.25%****	**0.64%*****	**0.49%****
Log dividend yield	−0.16%**	**0.31%****	**0.76%*****	**0.54%****
Log earning–price ratio	−1.38%*	**−0.45%****	**0.25%***	**0.30%***
Log smooth earning–price ratio	−0.79%*	**0.07%****	**0.48%****	**0.47%****
Log dividend–payout ratio	−1.57%	**−1.29%**	**−0.12%**	**0.03%**
Book-to-market ratio	−1.40%	**−0.88%**	**0.02%**	**0.04%**
T-Bill rate	0.01%*	**0.20%***	**0.10%**	**0.54%****
Long-term yield	−0.94%*	**0.20%****	**0.31%****	**0.55%*****
Long-term return	−0.74%	**−0.62%**	**0.19%**	**0.53%****
Term spread	0.06%	0.04%	0.05%	**0.24%***
Default yield spread	−0.19%	**−0.18%**	**0.18%**	**−0.08%**
Default return spread	−0.24%	−0.39%	**0.15%**	−0.30%
Stock variance	0.24%	0.11%	−0.17%	−0.29%
Net equity expansion	−0.79%	**−0.77%**	**−0.01%**	**−0.08%**
Inflation	−0.13%	−0.13%	**0.07%**	**−0.12%**
Log total net payout yield	−0.49%*	**0.09%***	**0.18%**	**−0.01%**
Average	−0.53%	**−0.22%**	**0.19%**	**0.18%**

Economic performance and portfolio choice

- So far we have compared the statistical performance of return forecasts
- Next evaluate the economic significance of these return forecasts by considering the optimal portfolio choice of an investor who uses the return forecasts.

Framework

Consider the optimal asset allocations

$$\omega_t^* = \arg\max_{\omega_t} \mathrm{E}[U(\omega_t, r_{t+1})|\mathcal{D}^t], \tag{26}$$

with $\mathcal{D}^t$ denoting all information available up to time t, and $t = \underline{t}-1, \ldots, \bar{t}-1$. The investor is assumed to have power utility

$$U(\omega_t, r_{t+1}) = \frac{[(1-\omega_t)\exp(r_{ft}) + \omega_t \exp(r_{ft} + r_{t+1})]^{1-A}}{1-A}. \tag{27}$$

Framework

Taking expectations with respect to the predictive density of r_t,

$$\omega_t^* = \arg\max_{\omega_t} \int U(\omega_t, r_{t+1}) p(r_{t+1}|\mathcal{D}^t)\, dr_{t+1}. \tag{28}$$

The integral in Eq. (28) can thus be approximated using the draws from the predictive densities as described in Section 2 and in the Appendix. Specifically, under the prevailing mean model, for suitably large values of J the solution to Eq. (28) can be approximated by

$$\widehat{\omega}_t = \arg\max_{\omega_t} \frac{1}{J} \sum_{j=1}^{J} \left\{ \frac{[(1-\omega_t)\exp(r_{ft}) + \omega_t \exp(r_{ft} + r_{t+1}^j)]^{1-A}}{1-A} \right\}. \tag{29}$$

Framework

Realized wealth at time $t+1$ under the prevailing mean model and the model based on predictor i:

$$\widehat{W}_{t+1} = (1-\widehat{\omega}_t)\exp(r_{ft}) + \widehat{\omega}_t \exp(r_{ft}+r_{t+1}),$$
$$\widehat{W}_{t+1,i} = (1-\widehat{\omega}_{t,i})\exp(r_{ft}) + \widehat{\omega}_{t,i} \exp(r_{ft}+r_{t+1}). \quad (31)$$

The certainty equivalent return for the model based on predictor i, CER_i, is defined as the value that equates the average realized utility of the prevailing mean model to the average realized utility of the model based on the ith predictor, over the forecast evaluation sample:

$$CER_i = \left[\frac{\sum_{\tau=\underline{t}}^{\bar{t}} \widehat{U}_{\tau,i}}{\sum_{\tau=\underline{t}}^{\bar{t}} \widehat{U}_{\tau}}\right]^{1/(1-A)} - 1, \quad (32)$$

Variable	Panel A: Monthly			
	No constraint	CT truncation	EP constraint	SR [0,1] constraint
Log dividend–price ratio	−0.28%	**−0.20%**	**0.32%**	**0.74%**
Log dividend yield	−0.26%	**−0.10%**	**0.41%**	**0.76%**
Log earning–price ratio	0.13%	**0.19%**	**0.49%**	**0.84%**
Log smooth earning–price ratio	−0.29%	**−0.18%**	**0.38%**	**0.81%**
Log dividend–payout ratio	0.20%	**0.23%**	**0.37%**	**0.70%**
Book-to-market ratio	−0.76%	**−0.58%**	**0.28%**	**0.63%**
T-Bill rate	−0.15%	**−0.14%**	**0.25%**	**1.06%**
Long-term yield	−0.26%	**−0.19%**	**0.25%**	**1.00%**
Long-term return	−0.03%	**0.01%**	**0.35%**	**1.41%**
Term spread	0.21%	0.20%	**0.29%**	**0.96%**
Default yield spread	−0.11%	**−0.10%**	**0.10%**	**0.56%**
Default return spread	−0.06%	−0.06%	**0.31%**	**0.65%**
Stock variance	0.17%	0.15%	−0.14%	**0.61%**
Net equity expansion	−0.05%	**−0.04%**	**0.25%**	**0.77%**
Inflation	−0.09%	−0.09%	**0.26%**	**0.72%**
Log total net payout yield	−0.30%	**−0.23%**	**0.30%**	**0.68%**
Percent equity issuing				
Average	−0.12%	**−0.07%**	**0.28%**	**0.81%**

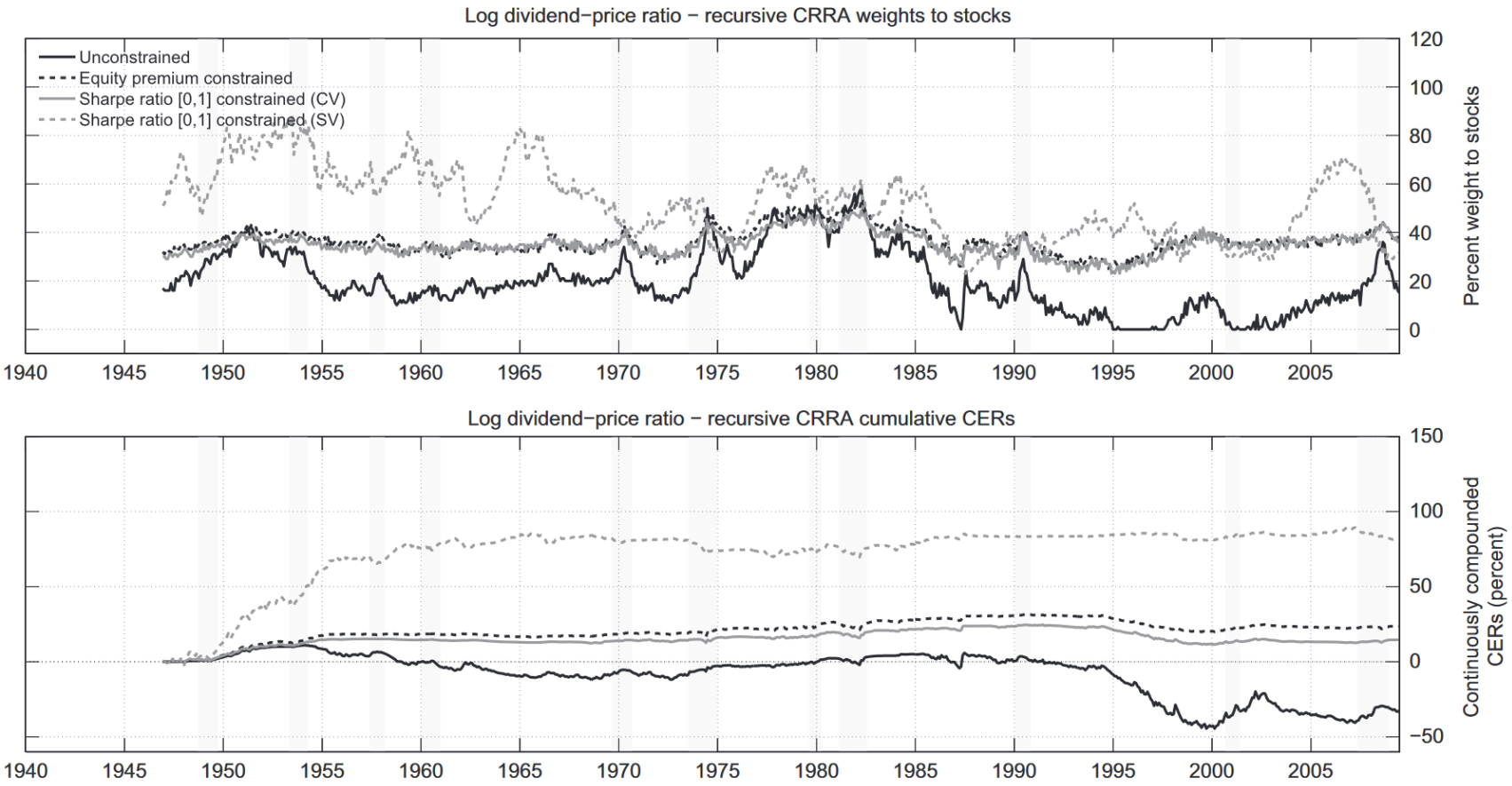

Fig. 14. Portfolio allocation and economic value of forecasts. This figure plots the percentage allocation to stocks and the resulting cumulative certainty equivalent returns (CERs) measured relative to the prevailing mean model. Each month we compute the optimal allocation to stocks and T-bills based on the predictive density of excess returns. The investor is assumed to have power utility with a coefficient of relative risk aversion of five and the weight on stocks is constrained to lie in the interval $[0, 0.99]$. The top graph shows the recursively computed optimal weight on stocks, and the bottom graph shows the cumulative certainty equivalent return measured relative to the prevailing mean model. The equity premium constrained model imposes the constraint that $\hat{r}_{\tau+1|t} = \int(\mu+\beta x_\tau)p(\mu,\beta|\mathcal{D}^t)\,d\mu\,d\beta > 0$, for $\tau = 1,\ldots,t$ and information set $\mathcal{D}^t$, and the Sharpe ratio constrained models impose that $0 \le \hat{r}_{\tau+1|t}/\hat{\sigma}_{\tau+1|t} \le 1$, for $\tau = 1,\ldots,t$, where $\hat{\sigma}_{\tau+1|t}$ is the posterior volatility estimate obtained from either a constant volatility model (CV) or a stochastic volatility model (SV). The predictor used is the log dividend–price ratio. CRRA: constant relative risk aversion.

Conclusion

- Economic constraints effectively shrink the predicted equity premium and reduce the effect of outliers.
- Imposing economic constraints on the equity premium forecast improves the predictive accuracy for nearly all of the prediction models.
- When used to select portfolio weights, the constrained forecasts are found to yield higher certainty equivalent returns than their unconstrained counterparts.

Thanks!